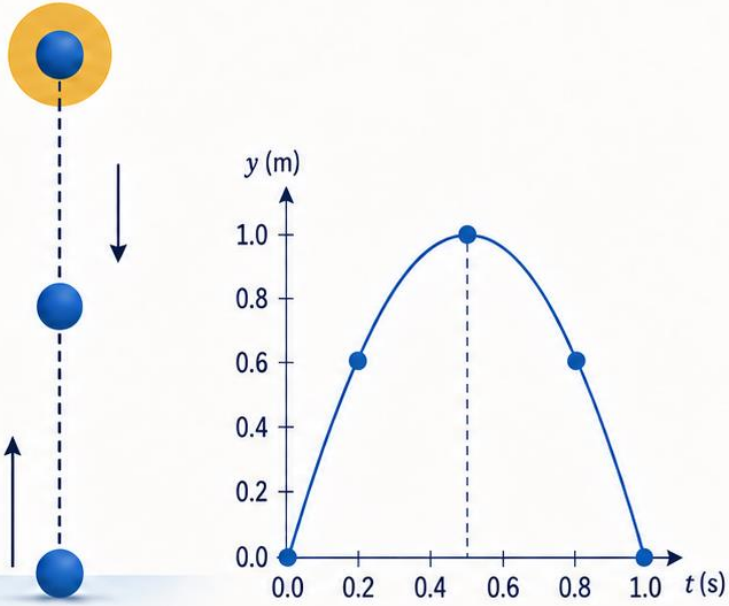
PHY4605 · WEEK 01 · LECTURE

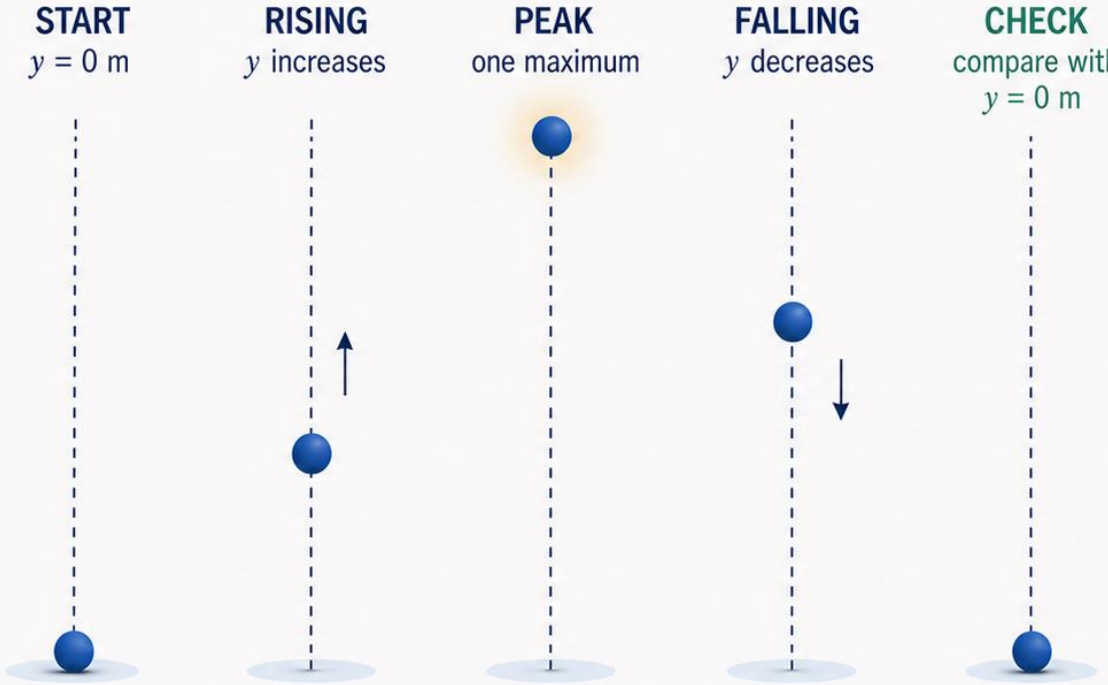
Arrays and Plots

Represent a physical model and compare prediction with evidence



Model → $\frac{m}{s^2}$ Units → Algorithm → </> Code → Validation → Interpretation

Before MATLAB, sketch what must happen



Prediction before computation

- At $t = 0$ s, $y = 0$ m
- As t increases, y rises to one maximum
- After the peak, y decreases
- Near the end, compare with $y = 0$ m

Prediction first

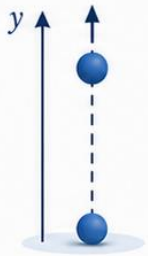
Sketch the shape before coding.

Decide what the computer must remember

- Scalars store fixed values such as y_0, v_0, and g.
- An array stores many chosen times.
- Every stored quantity needs a meaning and unit.

Physical model (vertical motion)

$$y(t) = y_0 + v_0\,t - \frac{1}{2}\,g\,t^2$$



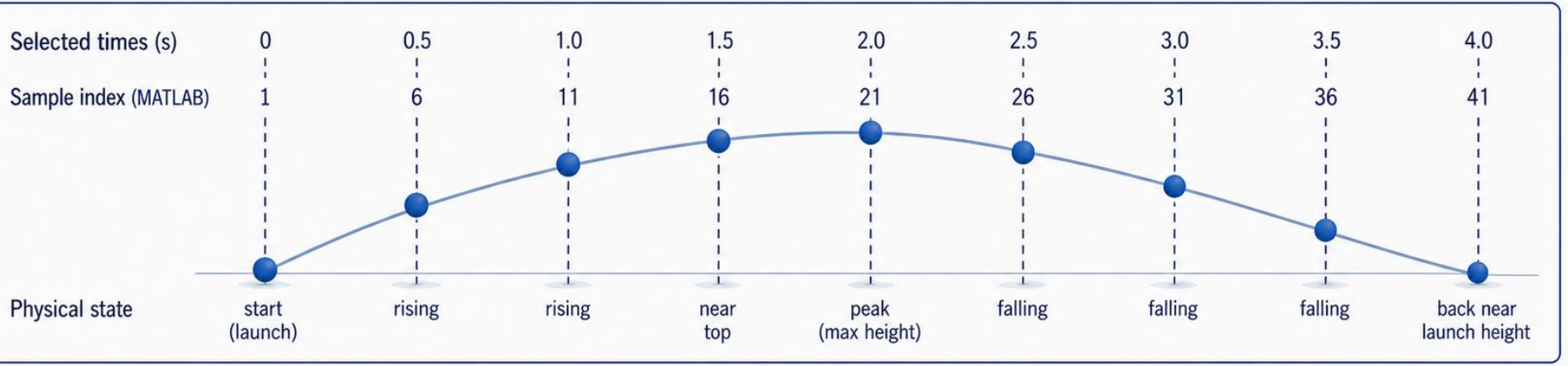
$y_0 = 0$ m
$v_0 = 20$ m s^{-1}
$g = 9.81$ m s^{-2}
$t = 0{:}0.1{:}4$ s

Upward is positive.
Air resistance is neglected.

	Quantity	MATLAB name	Storage form	Meaning	Unit
SCALARS (fixed values)	y_0	`y0_m`	scalar	initial height (launch height)	m
	v_0	`v0_mps`	scalar	initial upward speed	m s^{-1}
	g	`g_mps2`	scalar	acceleration due to gravity (positive number)	m s^{-2}
ARRAYS (many values)	t	`t_s`	$1 \times N$ array	chosen time samples	s
→	$y(t)$	`y_m`	$1 \times N$ array	height at each time in `t_s`	m

Blue = scalars (fixed values) | Green = arrays (many values) | Highlighted row = main output we will compute and plot

Sampling is a choice, not the motion itself



Array construction — Method 1

```
t_s = 0:0.1:4;
```

Endpoints: 0 s ● ——— ● 4 s

Array construction — Method 2

```
t_s = linspace(0,4,41);
```

Endpoints: 0 s ● ——— ● 4 s

41 samples

Key points

- `t_s = 0:0.1:4` and `linspace(0,4,41)` make 41 time samples.
- The endpoints and count match.
- Sampling changes what we see, not the underlying equation.

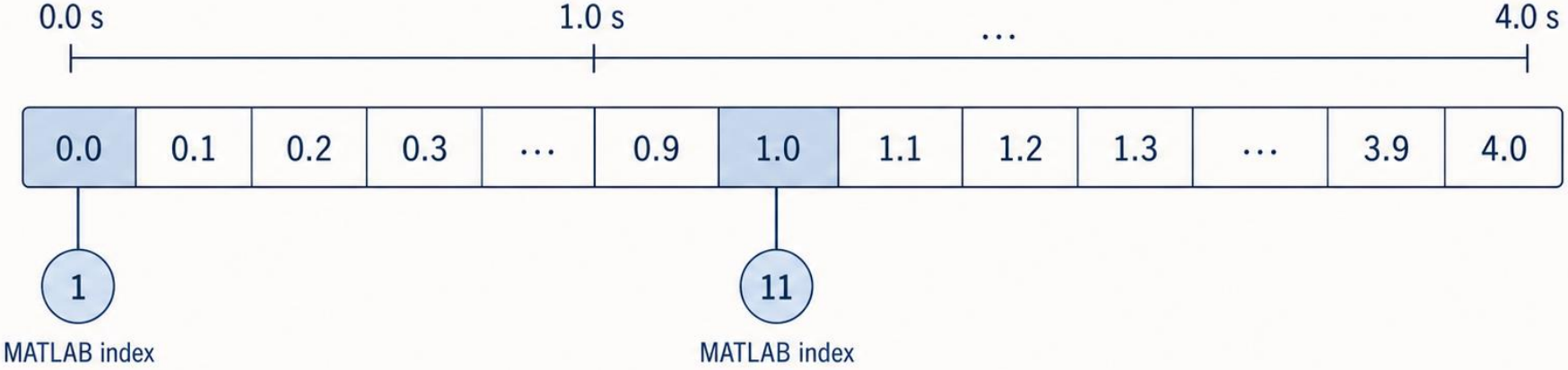
Model (continuous idea): $y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$

Representation (samples we use): selected moments from the same model

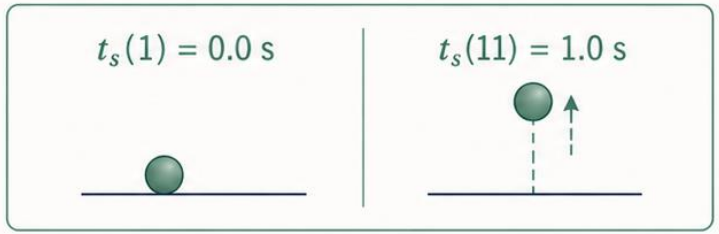
Different sampling → different detail, same underlying motion.

One index represents one physical moment

Time array: $t_s = 0:0.1:4\ \text{s}$



- $t_s(1)$ is the launch time.
- $t_s(11)$ selects 1.0 s.
- An index is a storage location, not a time value.



MATLAB indexing starts at 1

Make one equation work for many times

$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$$

starting position

$y_0 = 0$ m

launch contribution

$v_0 t \rightarrow$ metres

gravity contribution

$-\frac{1}{2} g t^2 \rightarrow$ metres

One stored time → one height

t_s	$[0.0$	0.1	0.2	$\ldots]$
y_m	$[0.0$	1.95	3.80	$\ldots]$

Each stored time gets one height.

Every term on the right has units of metres.

y_0 m | $v_0 t$ m | $\frac{1}{2} g t^2$ m

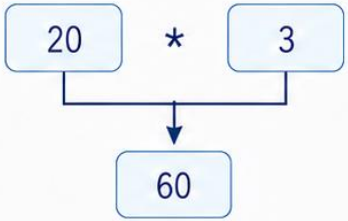
The missing dot changes the operation

The dot means apply the operation element by element.

Scalar multiplication

`20 * 3`

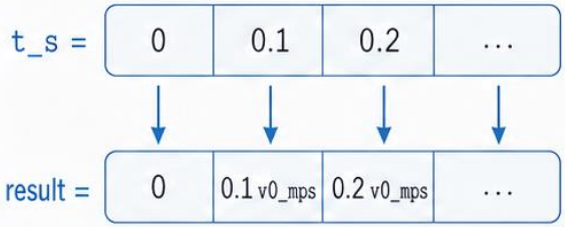
one scalar



`20 * 3` produces one scalar.

Scalar times array

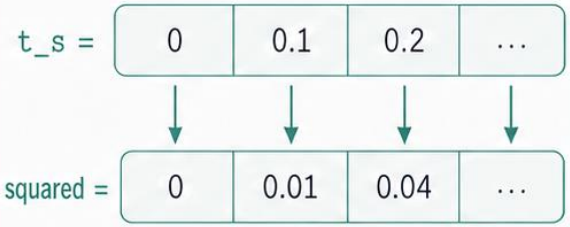
`v0_mps * t_s`



`v0_mps * t_s` produces one value per time.
Same result as `v0_mps .* t_s` when one operand is scalar — the dot is optional.

Square each stored time

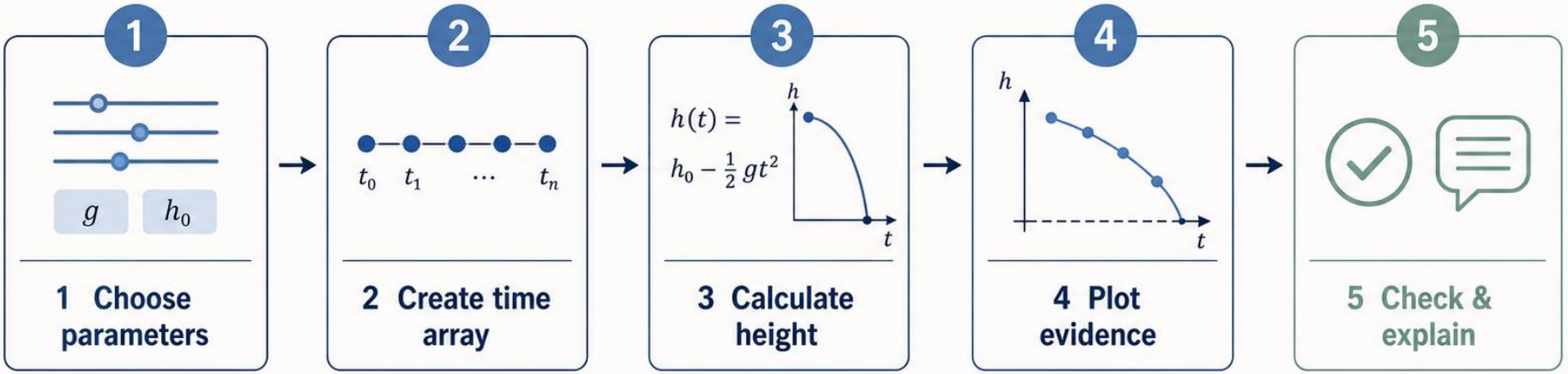
`t_s .* t_s`



`t_s .* t_s` squares every stored time value.
Dot required for array `.*` array.

Checkpoint: Which expression returns an array? ✓ `v0_mps * t_s` ✓ `t_s.^2`

Build the algorithm before writing syntax



- Choose the physical parameters.
- Create the time array.
- Calculate height at every time.
- Plot, check, and explain.

Read the Live Script in the order the physics runs

```
1  v0 = 20;
   g = 9.81;

2  t = linspace(0,4,11);

3  y = v0.*t - 0.5*g.*t.^2;

4  plot(t,y,'LineWidth',2);

5  y(1)
```

1 PARAMETERS

Set the launch speed and gravity.

2 ARRAY

Choose the sampled times.

3 CALCULATION

Evaluate height for every time.

4 PLOT

Turn the array into evidence.

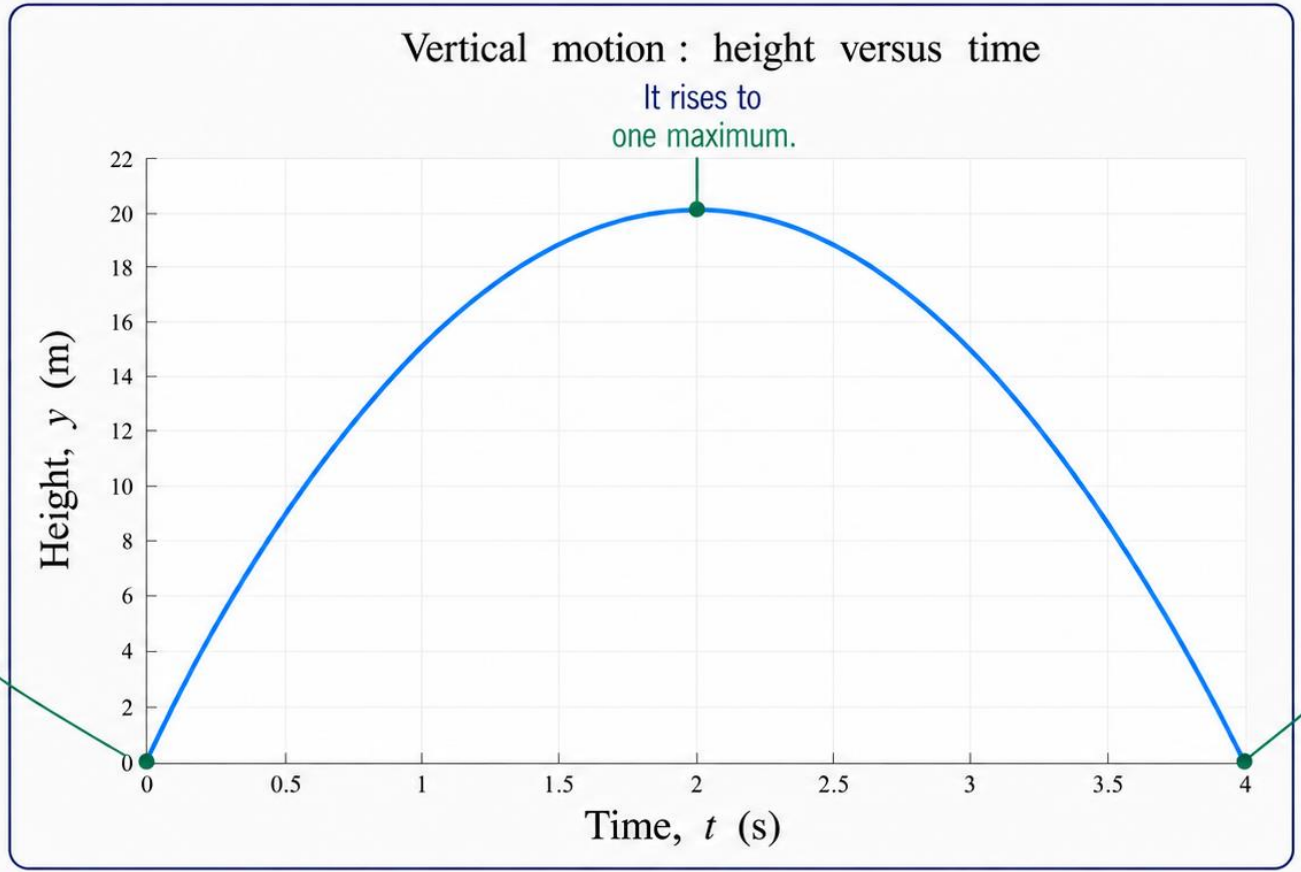
5 CHECK

At t = 0 s, y must be 0 m.

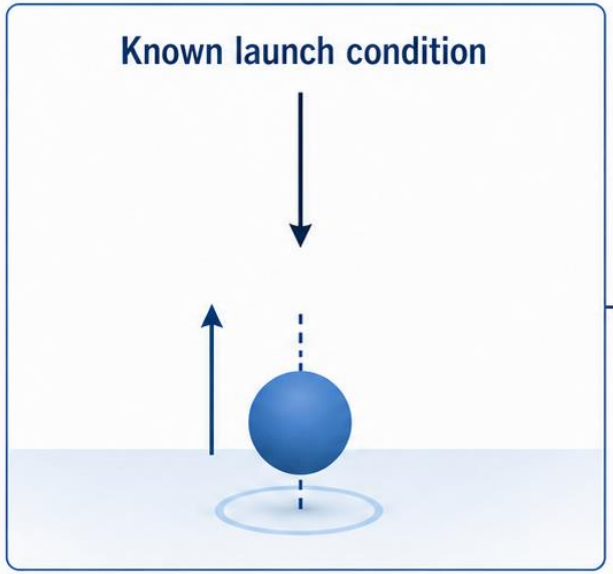
✓ expected: $y(1) = 0$

- Read comments and variable names first.
- Trace the array-ready model line.
- Locate the plot and boundary check.

Compare the prediction with the plotted evidence



Use a check the model must pass



- The known condition is $y(0) = 0$ m.
- In a one-based array this is $y_m(1)$.

first stored value

$y_m(1)$

$t = 0$ s

$y(1) = 0.000$ m

At $t = 0$ s, the model must give $y = 0$ m.

Executable boundary check

```
assert(abs(y_m(1)-y0_m) < 1e-12)
```

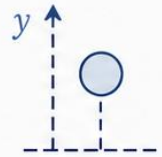
PASS — launch value matches $y0_m$

Why epsilon = 10^-12? Floating-point arithmetic may leave a tiny residual, so test close enough — not exactly zero.

A smooth graph is not enough: check a known condition.

State what the computation lets us claim

MODEL



$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$$

upward positive;
air resistance neglected

SAMPLING

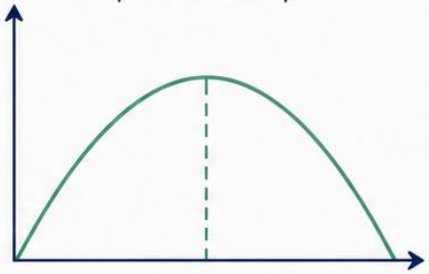


time array: 0:0.1:4 s

sampled positions

GRAPH

qualitative shape



rises to one maximum, then falls

CHECK



$y_m(1) = 0$ m

starting boundary condition

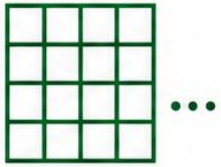
Under these assumptions, the ball rises to one maximum and the boundary check supports $y(0) = 0$ m.

Not supported: the model proves real motion exactly.

A result needs a unit, a trend, and a limitation.

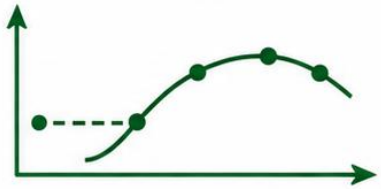
Takeaway and Next Step

1 CORE IDEA



A model becomes evidence when arrays carry physical values with units.

2 CHECK



Compare the $t = 0$ prediction with the first plotted value.

3 NEXT STEP



Use a visible physical recipe before MATLAB syntax.

? What changed?

? How do you know?